1 < 2 < 3

Variables
- numerical (can do arithmetic)
  - discrete — eg. numbers.
  - continuous — eg. weight, height.
    - A, B → always find blood pressure C, that A < C < B.
- categorical (cannot)
  - ordinal → eg. rankings
  - nominal → eg. colors, species, schools.

Graphical Display and Numerical Summary Worksheet

This worksheet is intended for week 3 of PSTAT 5LS. You do not need to turn this in.

Q1. Variable Type

For each of the following variables, classify them as either discrete, continuous, ordinal, or nominal, and use this to determine what type of visualization is best to plot the distribution of observations collected on each.

a. x = the place/ranking (first, second, third, etc.) of athletes at the end of a marathon

Ordinal variable.

b. y = the number of children in various families residing in the city of Santa Barbara

discrete variable.

2000. 2001.

↳ 2001.5

c. z = the species to which 100 different plants, selected from Leadbetter Beach, belong.

↳ nominal variable.

a. histograms. → distribution of one numerical variable.

b. bar charts. → one categorical variable.

c. scatter plots → relation of two numerical variables.

d. side-by-side boxplots. → one numerical variable across one categorical variable.

Q2. Which Plot?

Below, you are provided with the description of a particular dataset. Identify the type of visualization (e.g. histogram, scatterplot, etc.) that you believe would best achieve the stated goal, and provide a brief justification for your answer. Keep in mind that there are potentially multiple "correct" answers- as such, justify your answers!

a. An environmental scientist would like to see how (if at all) PM2.5 concentration (which is a measure of air quality) varies with temperature (as measured in Centigrade).

numerical vars.

Scatter plots

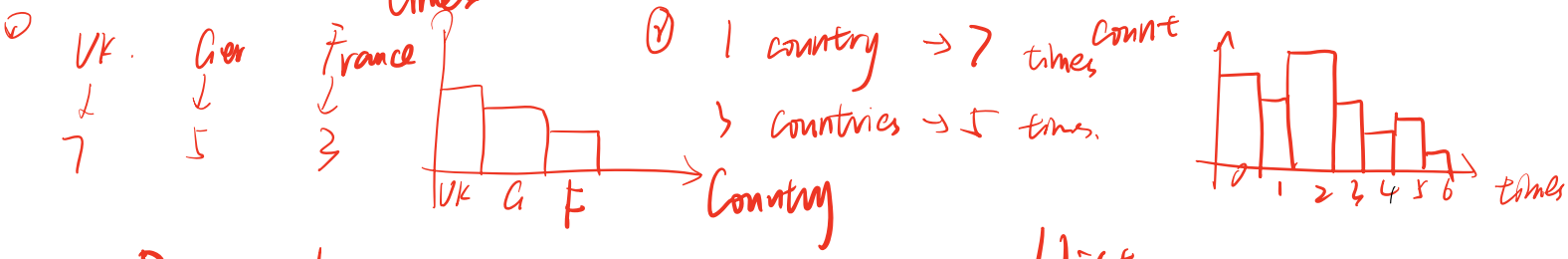
b. A clinical researcher has administered 4 different dosages of a particular medicine to a large set of volunteers, and would like to visualize how (if at all) the insulin levels of subjects varies across dosages.

categorical

numerical.

Side-by-side boxplots

c. An avid watcher of Eurovision has tallied up the number of times each country has won the competition, and would like to visualize their data.



Bar chart

Histogram

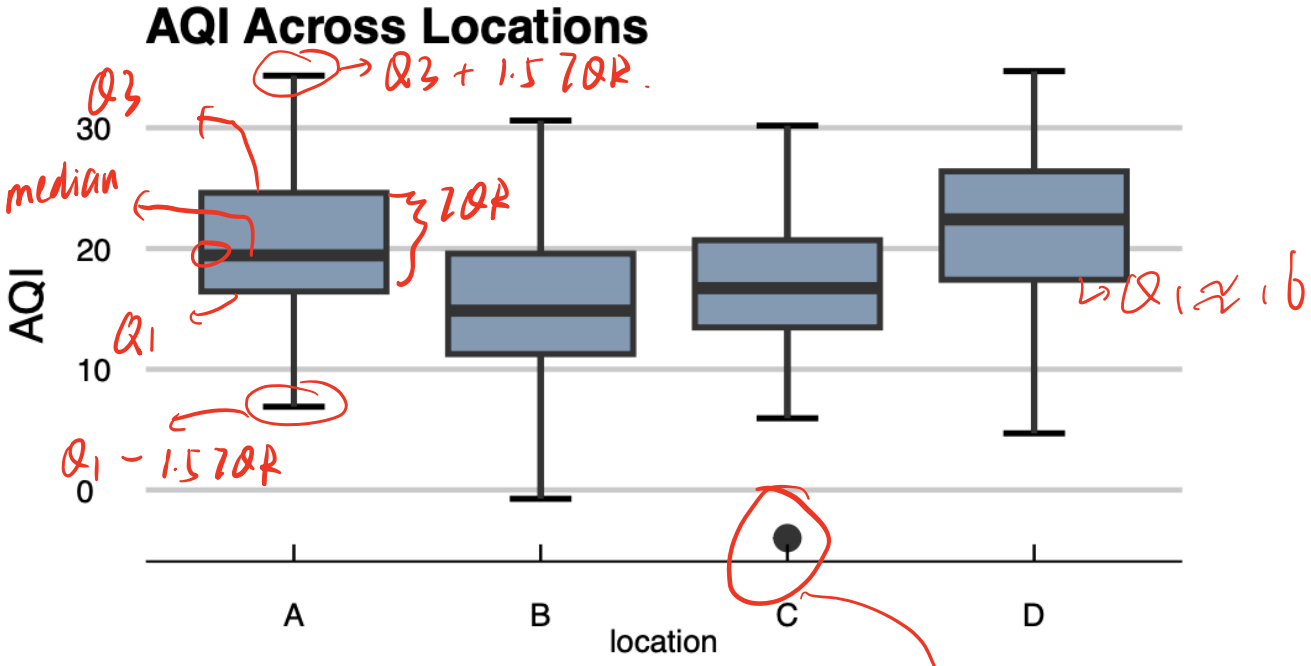
d. Alex has collected information on how long it takes a sample of 60 PSTAT 5LS students to complete the Final Exam, and would like to visualize the distribution of completion times.

num

histogram.

Q3. Boxplots

A researcher has collected measurements on the AQI (Air Quality Index) at several locations in Santa Barbara. She has labeled her locations A through D, and collected 100 AQI measurements from each location. The results of her study are displayed graphically below:



a. What is the (approximate) median AQI at location A?

20.

b. Approximately what percent of AQI readings at Location D were less than 16.

25%

c. Are there any outliers at any of the locations?

↳ points outside of the reach of whiskers.

d. From the plot, does it appear that the different locations have different AQI readings? Explain briefly.

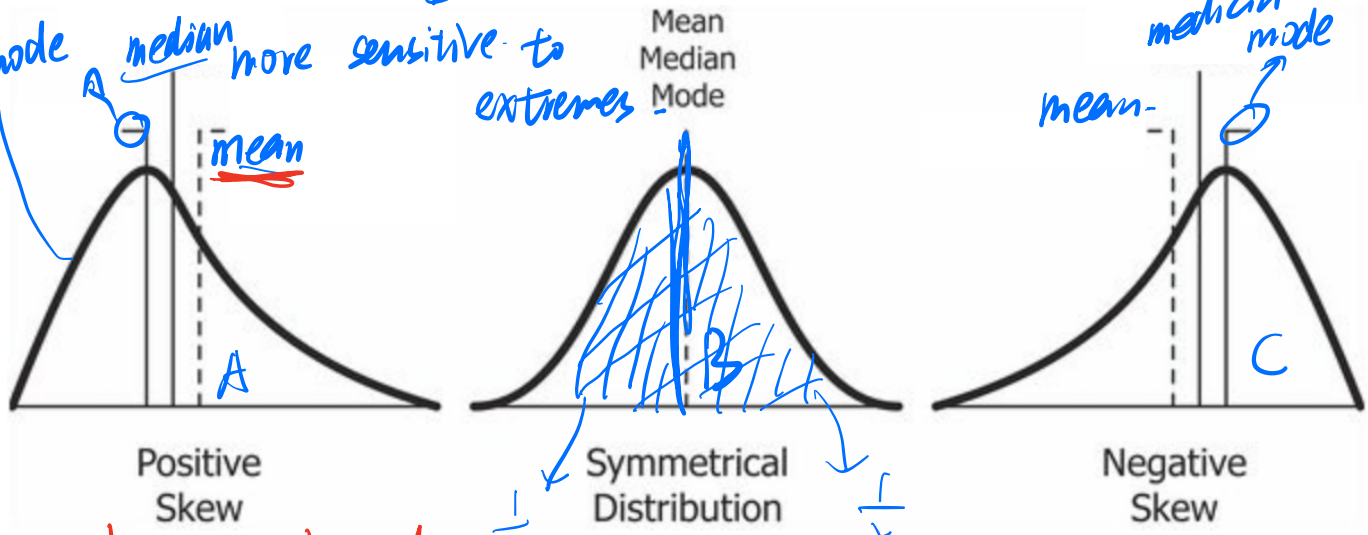
median

1 2 3 3.1 3.5 4 6 8 10

↳ divide area into two equal parts.

Q4. Skewness

Below is a plot of right skewed, symmetric, and left skewed distributions. Please fill in for the right and left skewed distribution where the mean, median, and mode are.



Q5. Parameter or Statistic?

Determine whether the provided quantity is a population parameter or a sample statistic. Please explain.

a. The proportion of trees in the Amazon rainforest that are mango trees.

b. The average AQI (Air Quality Index) of 25 randomly selected cities on August 1, 2022.

c. The longest amount of time a human can hold their breath underwater.

d. The median number of words of a sample of 100 randomly-selected books.